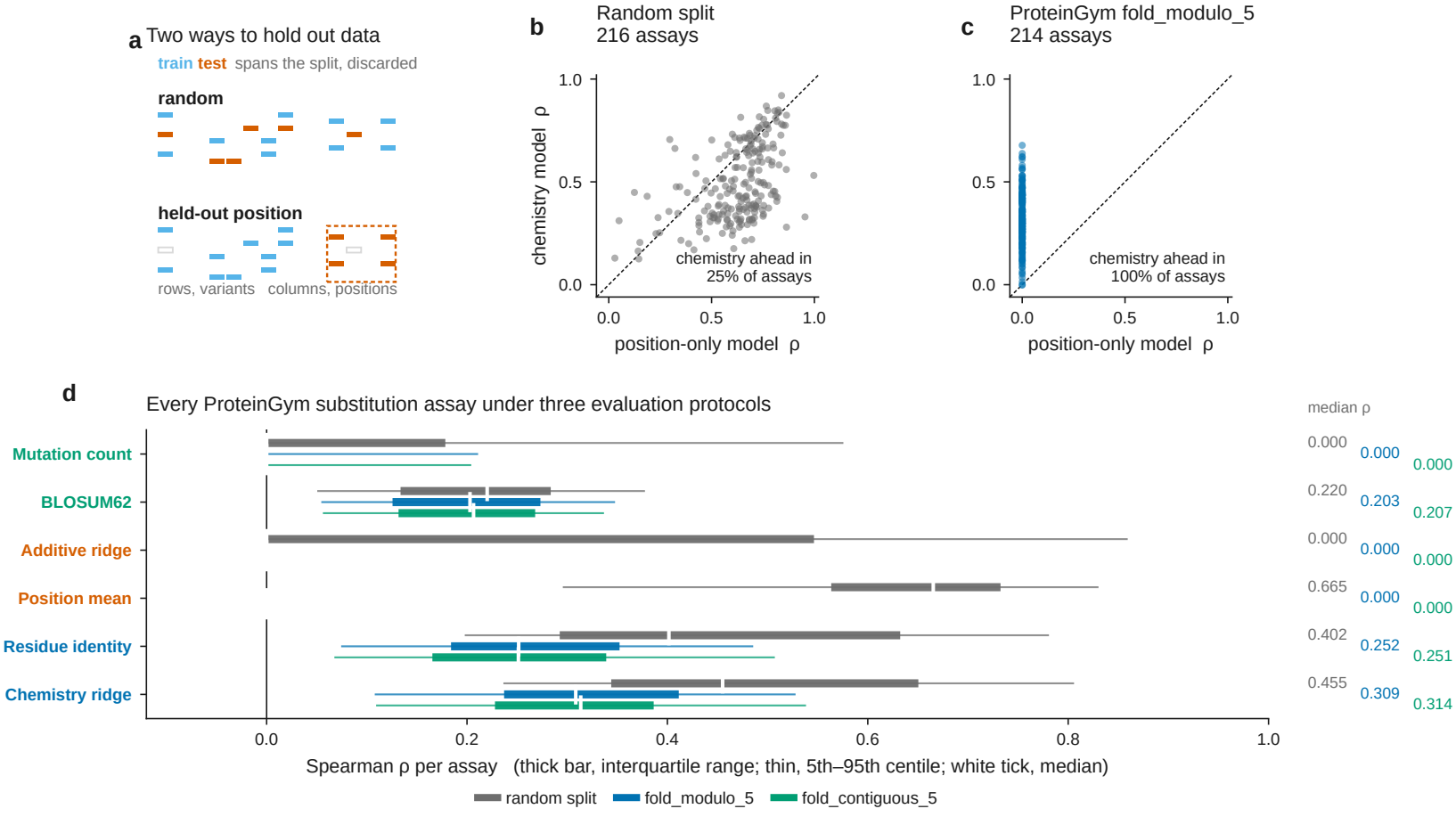
a
Two ways to hold out data
train test spans the split, discarded
random
held-out position
rows, variants
columns, positions
b
Random split
216 assays
chemistry model ρ
position-only model ρ
chemistry ahead in
25% of assays
c
ProteinGym fold_modulo_5
214 assays
position-only model ρ
chemistry ahead in
100% of assays
d
Every ProteinGym substitution assay under three evaluation protocols
median ρ
Mutation count
BLOSUM62
Additive ridge
Position mean
Residue identity
Chemistry ridge
0.000 0.000 0.000
0.220 0.203 0.207
0.000 0.000 0.000
0.665 0.000 0.000
0.402 0.252 0.251
0.455 0.309 0.314
Spearman ρ per assay (thick bar, interquartile range; thin, 5th–95th centile; white tick, median)
random split
fold_modulo_5
fold_contiguous_5